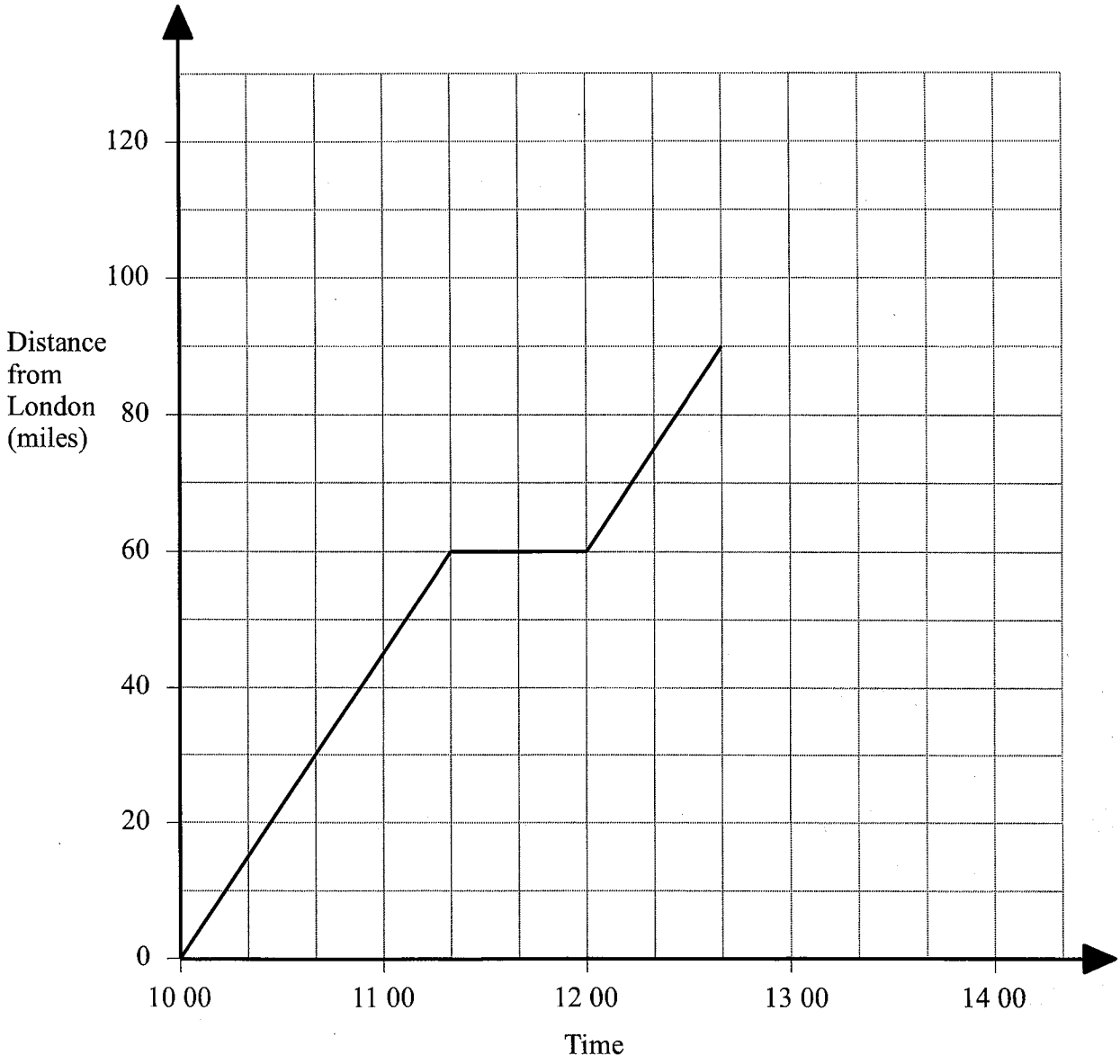
1 Dawn drove from London to Birmingham. She made one stop at a service station.

Here is part of Dawn's travel graph.



(a) For how many minutes did Dawn stop at the service station?

.....40.....minutes
(1)

(b) What was Dawn's average speed between London and the service station?

60 miles in 1hour 20 mins

$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{60}{4/3} = 45$ .....45.....miles/hour
(2)

(Total for Question 1 is 3 marks)

2 In a sale, the normal price of a car is reduced by 30%.
The sale price of the car is £5950

Work out the normal price of the car.

$$x \times 0.7 = 5950$$

$$x = \frac{5950}{0.7}$$

$$= 8500$$

£ 8500

(Total for Question 2 is 2 marks)

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3 (a) Write 3.71×10^{-5} as an ordinary number.

0.0000371

(1)

(b) Write 9 million in standard form.

9×10^{6}

(1)

(c) Calculate $(4.3 \times 10^{5}) \times (3.5 \times 10^{-2})$
Give your answer in standard form.

15050

1.505×10^{4}

(2)

(Total for Question 3 is 4 marks)

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4 Expand and simplify $(4x - 3)(x - 2)$

$4x^2 - 8x - 3x + 6$

$4x^2 - 11x + 6$

(Total for Question 4 is 2 marks)

5 Andrew ran 3.1 miles in 17 minutes and 25 seconds.

He assumes he can run 8 miles at the same speed.

Work out how long it would take Andrew to run 8 miles.

Give your answer in minutes and seconds to the nearest second.

17 mins 25 seconds $= \frac{209}{12}$ mins or $17\frac{5}{12}$ mins

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

$$= \frac{3.1}{209/12} = 0.178 \text{ miles/min}$$

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

$$= \frac{8}{0.178} = 44.9 \text{ mins} = 44 \text{ mins } 57 \text{ secs}$$

44 mins 57 secs

(Total for Question 5 is 4 marks)

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6 Solve $b^2 - 10b + 24 = 0$

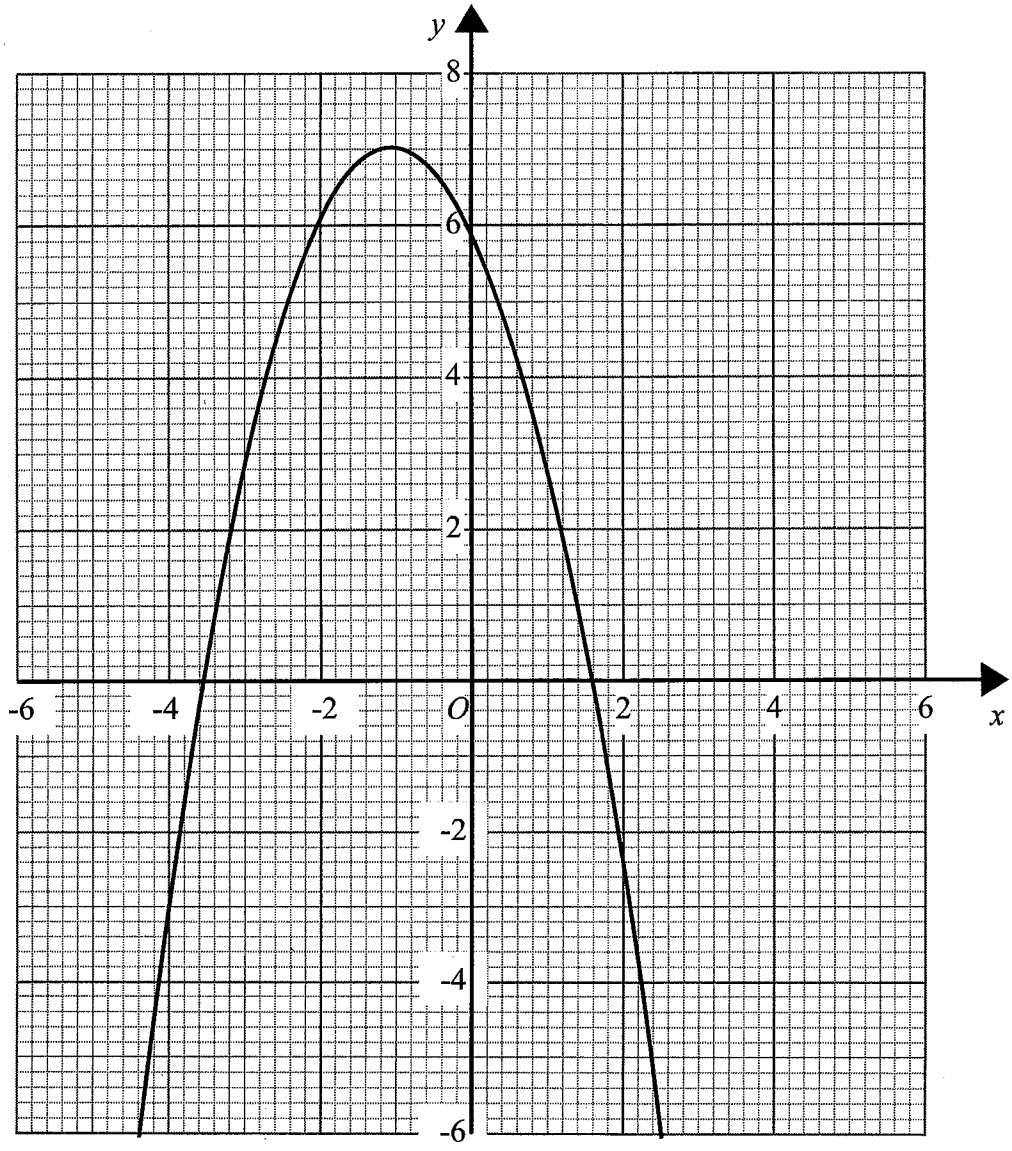
$(b-4)(b-6) = 0$

$b = 4 \qquad b = 6$

$b = 4$ or $b = 6$

(Total for Question 6 is 3 marks)

7 Here is the graph of $y = 6 - 2x - x^2$



(a) Write down the turning point of the graph $y = 6 - 2x - x^2$

(−1 , 7)

(1)

(b) Use the graph to find the roots of the equation $x^2 + 2x = 6$

1.6 and −3.6

(2)

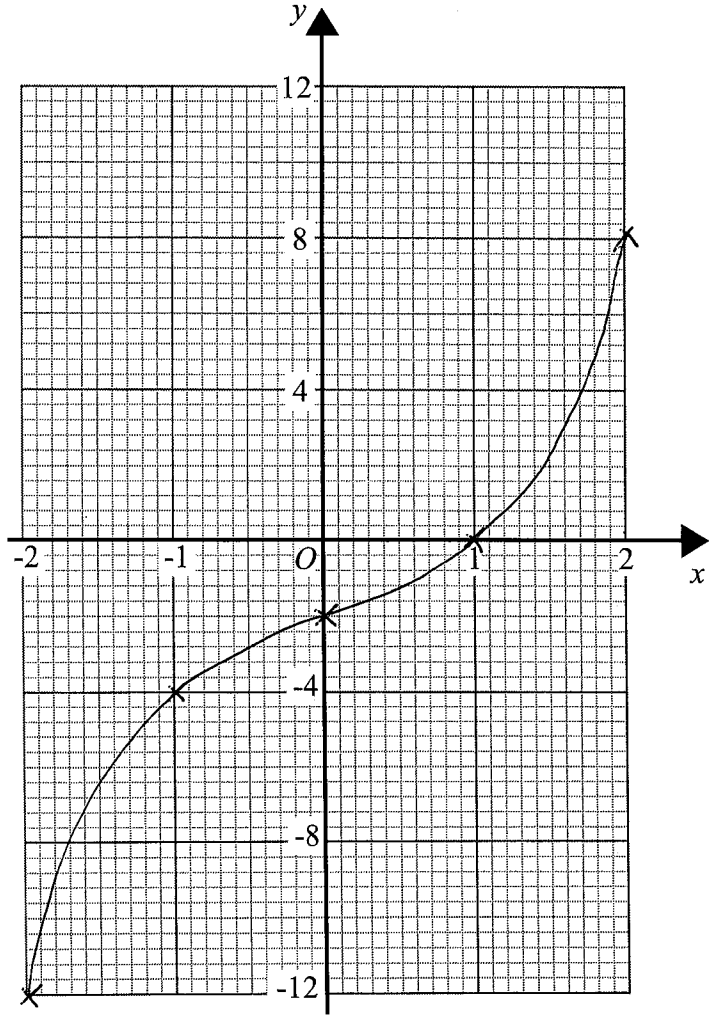
(Total for Question 7 is 3 marks)

8 (a) Complete the table of values for $y = x^3 + x - 2$

x	-2	-1	0	1	2
y	-12	-4	-2	0	8

(2)

(b) On the grid, draw the graph of $y = x^3 + x - 2$



(2)

(Total for Question 8 is 4 marks)